

ROGELIO (ROGER) LOZANO NAVA

Senior AI Architect & Full-Stack SaaS Engineer

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PROFESSIONAL SUMMARY

Systems Architect & AI Engineer fusing deterministic industrial logic (ISO 9001, Six Sigma) with SOTA probabilistic AI. Master in orchestrating cognitive frameworks (Python, MoE Routing, SQLite RAG) and deploying hyper-secure, \$0 CapEx B2B SaaS ecosystems (Vue 3, Nuxt 4 Nitro, Supabase). I deliver “Iron Truth” data pipelines, utilizing stochastic modeling to eliminate LLM hallucinations and rigorous engineering to bypass bottlenecks for high-growth tech ventures.

TECHNICAL SKILLS MATRIX

- **AI & Agentic Workflows:** Python, RAG (GraphRAG), Mixture of Experts (MoE Routing), MCTS, Multi-Agent Systems, Prompt Engineering, llama.cpp (Edge AI), Gemini 3.0 SDK.
- **Frontend & SaaS:** Vue 3, Nuxt 4 (Nitro), JavaScript, Tailwind CSS, Component-Driven Design, Edge Computing.
- **Backend & Cloud:** FastAPI, Serverless REST APIs, Cloudflare Pages/R2, Maileroo API, WebSockets, CI/CD Pipelines.
- **Data & Security:** PostgreSQL, Supabase, Row Level Security (RLS), Multi-Tenant Isolation, IAM, SQLite.
- **Data Eng & OSINT:** Map-Reduce OSINT Pipelines, XML/JSON Structuring, ArXiv/GitHub Extraction, Pandas, NumPy.
- **Industrial Logic:** Six Sigma (DMAIC), Monte Carlo Simulations, BPMN, ISO 9001 Governance, \$0 CapEx Architecture.

FLAGSHIP ARCHITECTURAL PROJECTS

Creator & Principal AI Architect | VROMLIX Prime (Cognitive Edge Architecture)

- **Reduced retrieval latency 99% and eliminated context amnesia** via a deterministic Python architecture, upgrading standard RAG to local GraphRAG (1-Hop) using SQLite and Hindsight W-B-O-S memory.
- **Established a “Zero-Hallucination” baseline**, generating 100% deterministic XML knowledge graphs by fusing ISO 9001 logic with Monte Carlo Tree Search (MCTS) to simulate and score MoE Routing routing paths pre-execution.
- **Secured autonomous code generation and file mutations** via a Dual-LLM “Maker-Checker” adversarial architecture, pairing a Surgeon agent with an isolated Forensic Auditor behind a strict HitL Sandbox Firewall.
- **Stabilized high-entropy pipelines and bypassed API limits** (30k TPM) using an Active Key Manager with 61-second Round-Robin cooldowns and exponential backoff, ensuring zero failures during parallel swarm execution.
- **Deployed enterprise AI on constrained edge hardware** (8GB RAM) with 0 OOM errors, implementing 8-bit KV Cache quantization and dynamic context compression via `llama-cpp-python`.
- **Automated SOTA technical intelligence gathering** via custom OSINT Map-Reduce scripts that autonomously scrape, filter (10+ GitHub stars threshold), and synthesize ArXiv papers and RSS feeds.
- **Validated system intelligence against industry standards** (SWE-bench, GAIA) by building a headless FastAPI proxy server for autonomous machine-to-machine evaluation.

Lead Software Architect | NexoContable (B2B2C Serverless SaaS Platform)

Live MVP: nexocontable.rlozano-intel.workers.dev

- **Achieved \$0.00 initial CapEx scalability and architected a B2B2C monetization engine** by deploying a multi-tenant SaaS on a serverless edge stack (Vue 3, Nuxt 4 Nitro, Cloudflare Pages/R2), integrating Stripe Checkout for recurring subscriptions.
- **Engineered enterprise data isolation** with zero cross-tenant vulnerabilities via complex PostgreSQL Row Level Security (RLS) and IAM within Supabase.
- **Automated transactional communications** via a decoupled serverless notification pipeline (Maileroo API) synchronized with real-time UI reactivity (Supabase WebSockets).

Creator & Lead Engineer | Sigma-Zero (Industrial Quality AI Orchestrator)

Public Repository: github.com/rlozanointel/sigma-zero

- **Automated DMAIC Define, Measure & Analyze phases** via a deterministic Python orchestrator calculating Process Capability (Cpk) and generating Pareto visualizations from defect logs, eliminating manual analysis.
- **Achieved zero-hallucination root cause synthesis** by anchoring RAG exclusively to a local sqlite-vec knowledge base, grounding failure causes in validated historical evidence to eliminate speculation.
- **Delivered a zero-configuration demo** executable in <30s via strict Pydantic v2 schemas, full pytest coverage, and a reproducible `./demo.sh` pipeline.

PROFESSIONAL EXPERIENCE

Processes Coordinator | *Grupo CINTE* / Mar 2014 – Mar 2016

- **Achieved zero non-conformities in international audits** by deploying an ISO 9001 Quality Management System, translating probabilistic workflows into deterministic BPMN architectures.
- **Increased resource allocation efficiency** by leading Business Process Management (BPM) governance aligned with PCF-APQC standards.

Process Optimization Lead | *ERP Soluciones* / Jan 2013 – Dec 2013

- **Ensured on-time Oracle EBS re-implementation** by standardizing enterprise processes and streamlining legacy workflows for a major transportation client.

Process & Quality Analyst | *State Government* / Mar 2009 – Sep 2013

- **Executed 3,000+ annual consulting sessions** enforcing strict ISO 9001 compliance via comprehensive documentation reviews.
- **Improved operational throughput** and eliminated bottlenecks by implementing statistical evidence-driven corrective actions.

Procedures Analyst | *Banca Mifel* / Sep 2006 – Feb 2008

- **Maintained zero regulatory penalties** (CNBV compliance) by optimizing and standardizing critical banking risk-control procedures.

EDUCATION & CERTIFICATIONS

- **MEng, Quality Systems & Productivity (Partial)** | Tecnológico de Monterrey
- **BSc, Industrial Engineering** | Instituto Tecnológico de Zacatepec
- **Internal Quality Auditor (ISO 9001)** | SINCAL
- **Google Data Analytics Certificate** | Coursera